

EMPLOYEE ATTRITION & WORKFORCE PERFORMANCE ANALYSIS

TechCrush Corporation

IBM HR Analytics Dataset — 1,470 Employees

Prepared for:

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EXECUTIVE SUMMARY

This report presents a comprehensive data-driven analysis of employee attrition and workforce performance at TechCrush Corporation, using the IBM HR Analytics dataset comprising 1,470 employees across three departments and nine job roles. The analysis was conducted to support leadership in making informed decisions on talent retention, compensation planning, training investments, and organisational culture improvements.

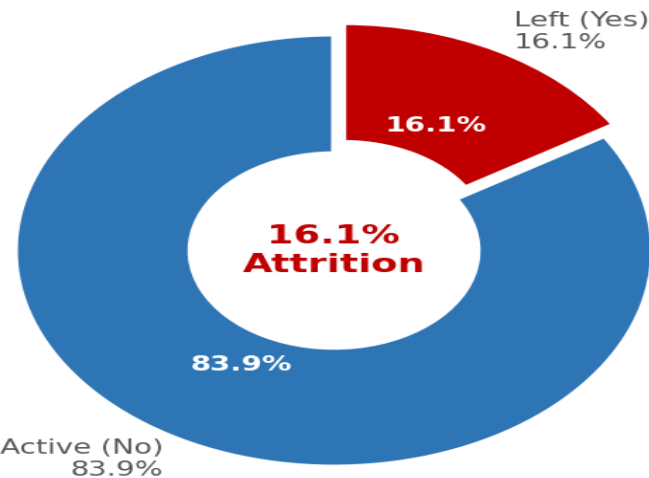
The findings reveal an overall attrition rate of 16.1%. This indicates a retention challenge that requires urgent strategic intervention. The analysis identifies five main drivers of attrition: overtime workload, business travel frequency, below-average compensation, absence of stock options, and low job satisfaction scores.

The most at-risk employee segments are Sales Representatives (39.8% attrition), Laboratory Technicians (23.9%), and Human Resources staff (23.1%). Employees who work overtime leave at nearly three times the rate of those who do not (30.5% vs 10.4%), and frequent travellers leave at more than three times the rate of non-travellers (24.9% vs 8.0%). Compensation data shows that employees who left earned an average of \$4,787/month compared to \$6,833 for those who stayed which gave a gap of over \$2,000.

The Key recommendations to these findings include; reducing overtime dependency, extending stock options to entry-level staff, closing income gaps for frontline roles, investing in employee training, and creating transparent career progression pathways.

Question1: OVERALL EMPLOYEE ATTRITION RATE

Q1 - Overall Employee Attrition Rate
(1,470 Employees)



Findings

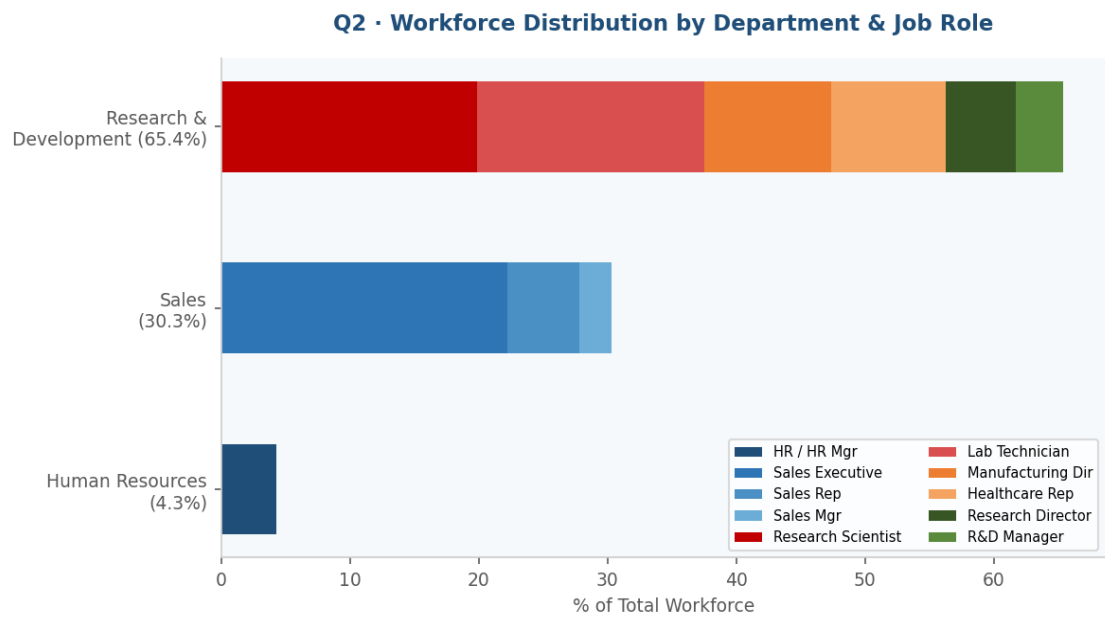
Attrition Status	Number of Employees	Percentage
Active (No)	1,233	83.9%
Left (Yes)	237	16.1%
Total	1,470	100%

Out of 1,470 employees at TechCrush Corporation, 237 employees left the organisation, producing an overall attrition rate of 16.1%. The remaining 1,233 employees (83.9%) are still active.

Insight: An attrition rate of 16.1%. This means that for every 100 employees, approximately 16 leave annually, this can be costly in terms of recruitment, onboarding, lost institutional knowledge, and reduced team productivity. If not addressed, TechCrush risks a compounding talent drain that will affect delivery capacity and business growth.

Recommendation: Treat this attrition rate as a business-critical metric. Leadership should set a target to reduce it within 12–18 months and track it quarterly as a KPI.

Question 2: Workforce Distribution by Department and Job Role



Findings

Department	Job Role	% of Total Workforce
Research & Development (65.4%)	Research Scientist	19.9%
Research & Development	Laboratory Technician	17.6%
Research & Development	Manufacturing Director	9.9%
Research & Development	Healthcare Representative	8.9%
Research & Development	Research Director	5.4%
Research & Development	Manager	3.7%
Sales (30.3%)	Sales Executive	22.2%
Sales	Sales Representative	5.6%
Sales	Manager	2.5%
Human Resources (4.3%)	Human Resources	3.5%
Human Resources	Manager	0.7%

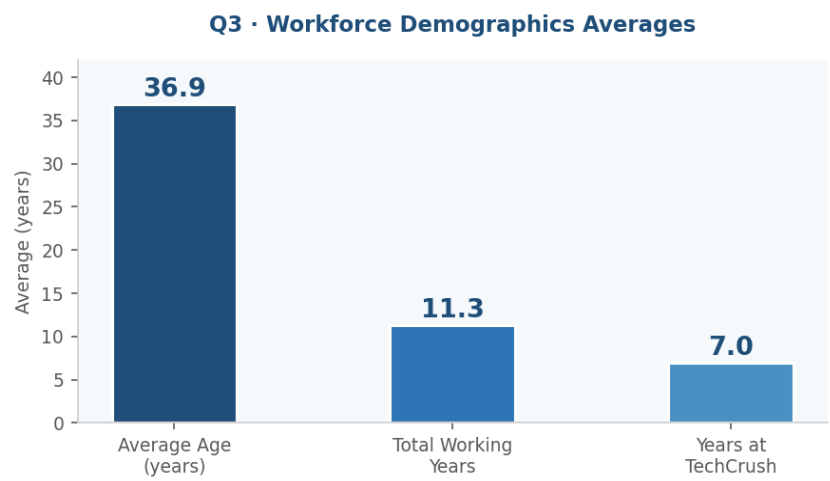
Research & Development is the dominant department at 65.4% of total headcount. Sales accounts for 30.3%, while Human Resources is the smallest at 4.3%. Within R&D, Research Scientists and Laboratory Technicians form the majority of percentage workforce. In Sales, the Sales Executive role alone accounts for 22.2% of the entire company workforce.

Insight:

The Research & Development (R&D) department has the largest share of employees in the organization, making it the primary workforce of the organisation. The Sales department has fewer employees than R&D but is largely made up of Sales Executives, this shows that this role is critical to its operations. In contrast, the Human Resources department has the smallest workforce, reflecting a little support function compared to other departments.

Recommendation: Techcrush should regularly review employee distribution across departments to ensure that each area has adequate staffing to meet its responsibilities.

Question 3: Average Age, Total Working Years, and Years at Company



Findings

Metric	Average Value
Average Age	36.9 years
Average Total Working Years	11.3 years
Average Years at TechCrush	7.0 years

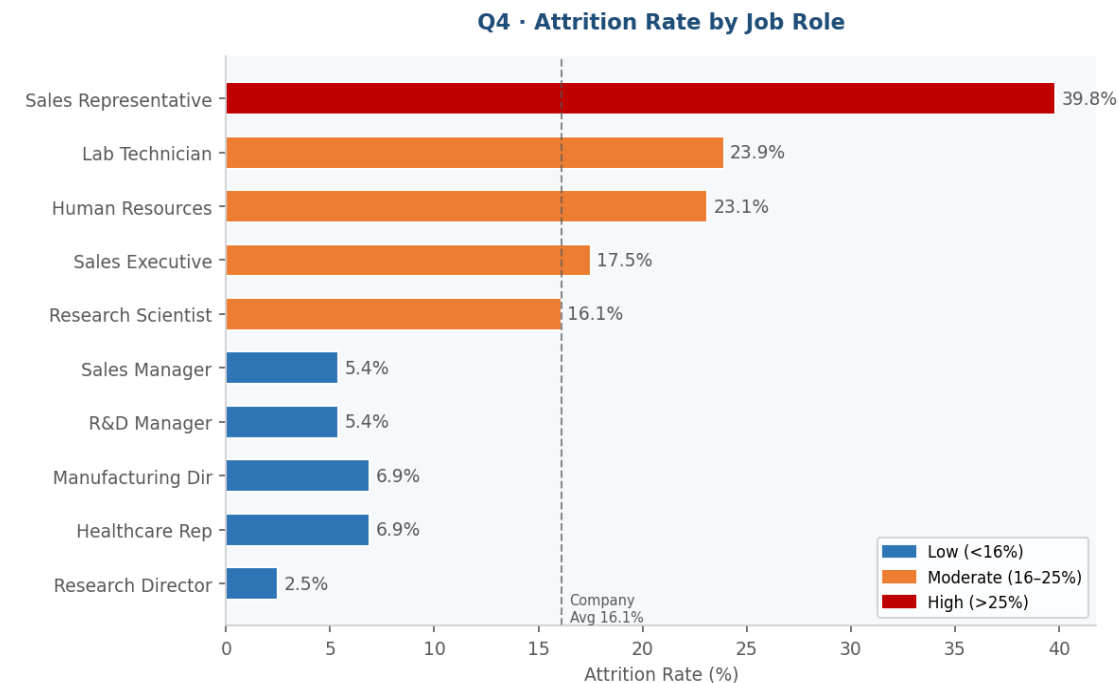
The average TechCrush employee is approximately 37 years old with over 11 years of total career experience, of which 7 years have been spent at TechCrush. This means employees arrive with roughly 4 years of prior experience before joining.

Insight:

The workforce at TechCrush is largely made up of experienced, mid-career employees, with an average of over 11 years of total work experience. This suggests that the company benefits from a knowledgeable and skilled workforce that has accumulated relevant professional expertise over time.

Recommendation: TechCrush should focus on retaining and developing its experienced employees through career growth opportunities, employee engagement initiatives, and competitive rewards to preserve valuable knowledge and expertise within the organization.

Question 4: Attrition by Department and Job Role



Findings

Job Role	Department	Attrition Rate
Sales Representative	Sales	39.8%
Laboratory Technician	R&D	23.9%
Human Resources	HR	23.1%
Sales Executive	Sales	17.5%
Research Scientist	R&D	16.1%
Manufacturing Director	R&D	6.9%
Healthcare Representative	R&D	6.9%
Manager (all depts)	All	~5.4%
Research Director	R&D	2.5%

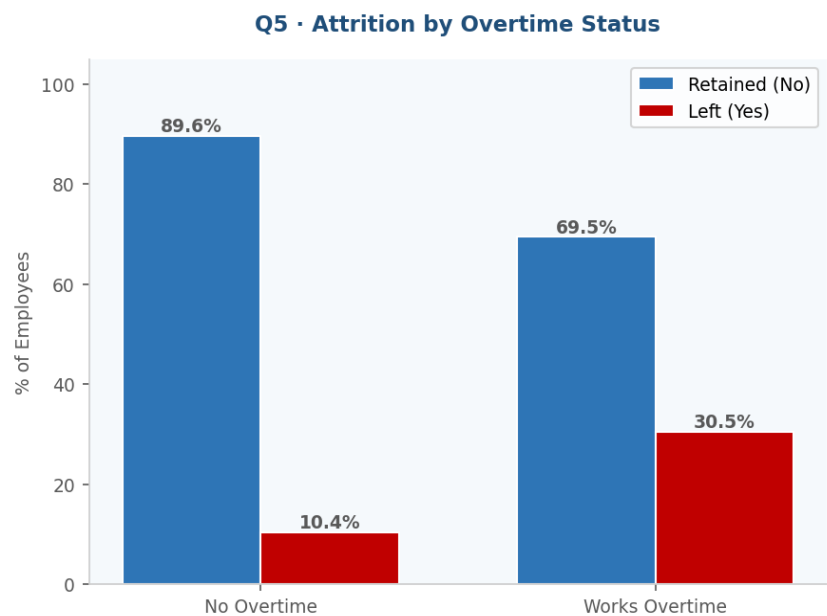
Sales Representatives have the highest attrition at 39.8% — nearly 2 in 5 leave each year. Laboratory Technicians (23.9%) and HR staff (23.1%) follow. At the other extreme, Research Directors (2.5%) and Managers (~5.4%) show the strongest retention. The Sales department overall has 20.6% attrition, R&D has 13.8%, and HR has 19%.

Insight:

Employee turnover is highest among Sales Representatives, Laboratory Technicians, and HR staff, while Managers and Research Directors have the lowest turnover rates. This suggests that employees in junior and operational roles are more likely to leave the company than those in senior positions.

Recommendation: The company should focus on improving retention among Sales Representatives, Laboratory Technicians, and HR staff by identifying and addressing the factors that contribute to their higher attrition rates.

Question 5: Attrition Among Overtime vs Non-Overtime Employees



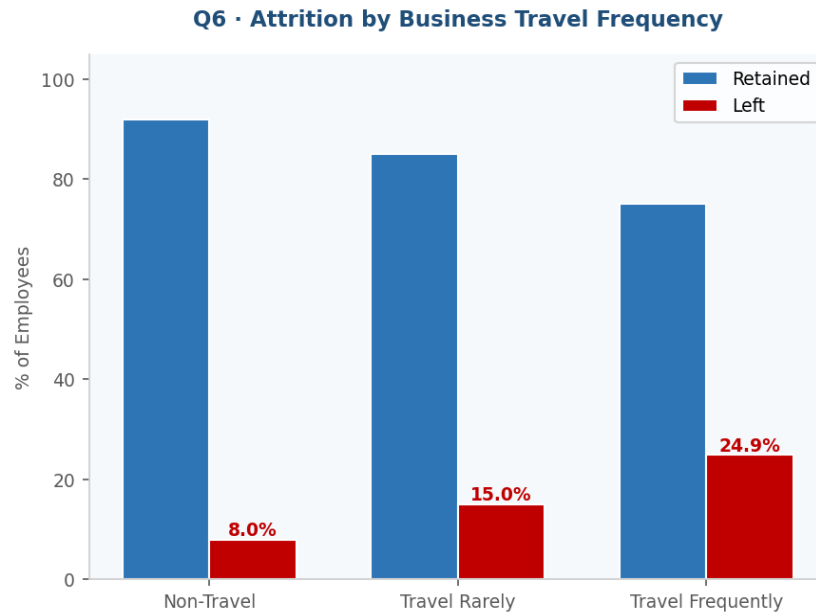
Overtime Status	Attrition Rate	Retention Rate
Works Overtime	30.5%	69.5%
Does Not Work Overtime	10.4%	89.6%

Employees who work overtime have an attrition rate of 30.5%, compared to just 10.4% for those who do not. Overtime employees are nearly three times more likely to leave TechCrush.

Insight:
Employees who work overtime are much more likely to leave the company than those who do not. This suggests that heavy workloads and poor work-life balance may be contributing to employee turnover. This is especially dangerous if overtime is concentrated in already high-attrition roles like Sales Representatives and Laboratory Technicians.

Recommendation: Check which departments carry the highest overtime burden. Redistribute workloads, consider additional hiring, or introduce process automation to reduce highest overtime. Reduce overtime hours and offer compensatory time off.

Question 6: Business Travel Frequency and Attrition



Findings

Travel Frequency	Attrition Rate	Retention Rate
Travel Frequently	24.9%	75.1%
Travel Rarely	15.0%	85.0%
Non-Travel	8.0%	92.0%

There is a clear progressive relationship between travel frequency and attrition. Frequent travellers have a 24.9% attrition rate, compared to 15.0% for rare travellers and just 8.0% for non-travellers. Frequent travellers are more than three times as likely to leave as non-travellers.

Insight:

Employees who travel more tend to leave more often. Attrition increases from 8% (no travel) to 15% (rare travel) and 24.9% (frequent travel) this shows a clear pattern.

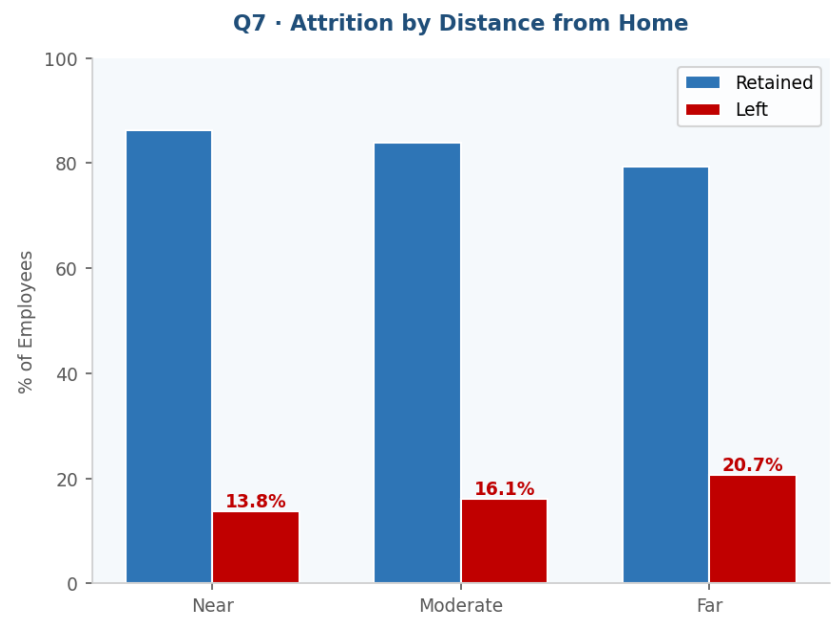
Frequent travel likely affects work-life balance, family time, and health, making employees more likely to quit.

TechCrush should consider reducing unnecessary travel, especially in Sales, and use virtual meetings where possible.

Recommendation: Techcrush should limit how often employees can travel each quarter, especially in high-travel roles.

Also, use online meetings as the main way to meet clients, so less physical travel is needed.

Question 7: Distance from Home and Attrition



Findings

Distance Group	Attrition Rate	Retention Rate
Far (>20 miles)	20.7%	79.3%
Moderate (10–20 miles)	16.1%	83.9%
Near (<10 miles)	13.8%	86.2%

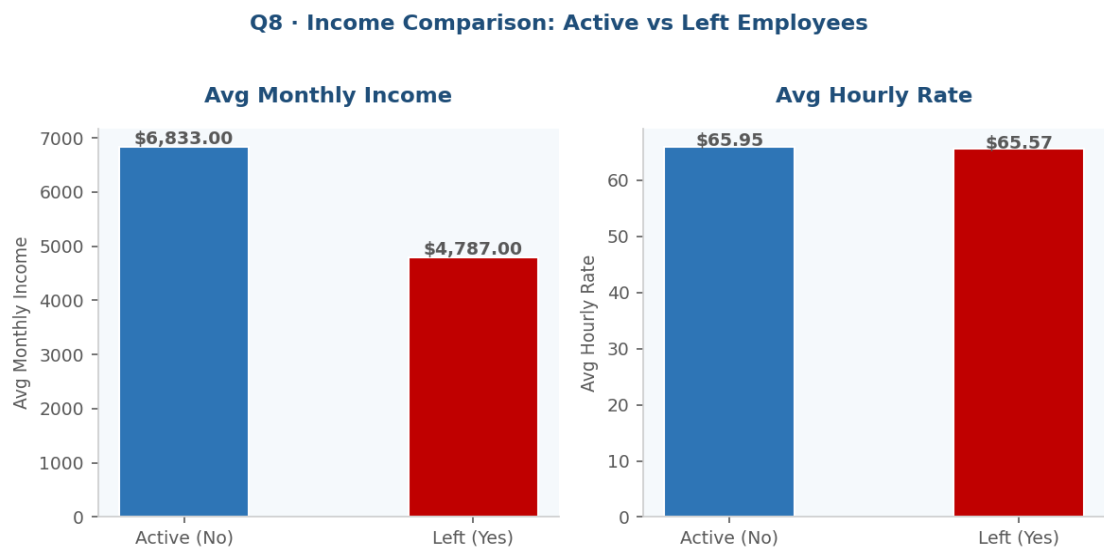
Employees living far from the office have a noticeably higher attrition rate (20.7%) than those living near (13.8%). The moderate distance group at 16.1% aligns with the company overall average.

Insight:

Employees who live farther away are more likely to leave than those who live nearby, with a noticeable difference of about 7%. Long travel reduce personal time, increase tiredness, and cost more money, which can make employees less likely to stay.

Recommendation: Implement a hybrid work policy or provide transport allowances for employees in the far distance category, particularly for roles where remote work is feasible.

Question 8: Monthly Income and Hourly Rate by Attrition Status



Findings

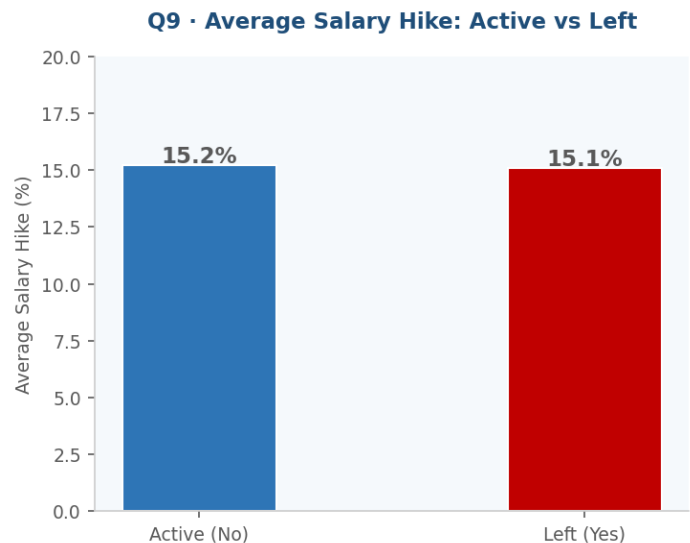
Attrition Status	Avg Monthly Income	Avg Hourly Rate
Active (No)	\$6,833	\$65.95
Left (Yes)	\$4,787	\$65.57
Overall Average	\$6,503	\$65.89

Employees who left earned an average monthly income of \$4,787 versus \$6,833 for those who stayed with a gap of \$2,046/month or \$24,552 annually. However, hourly rates are almost identical between both groups (\$65.57 vs \$65.95), indicating the income gap is driven by job level and role type, not hourly rate differences.

Employees who leave the company are mostly in lower-paying roles, This is in alignment with Q4 findings on Sales Representatives and Laboratory Technicians. Even though hourly pay is similar across groups, monthly earnings and total benefits matter too. This shows that lower-income employees are leaving because their pay is not competitive enough.

Recommendation: Compare salaries for Sales Representatives, Laboratory Technicians, and HR staff with what other companies pay. Then add extra monthly pay or performance bonuses for entry-level to mid-level workers.

Question 9: Percent Salary Hike and Employee Attrition



Findings

Attrition Status	Average Salary Hike
Active (No)	15.2%
Left (Yes)	15.1%
Overall Average	15.2%

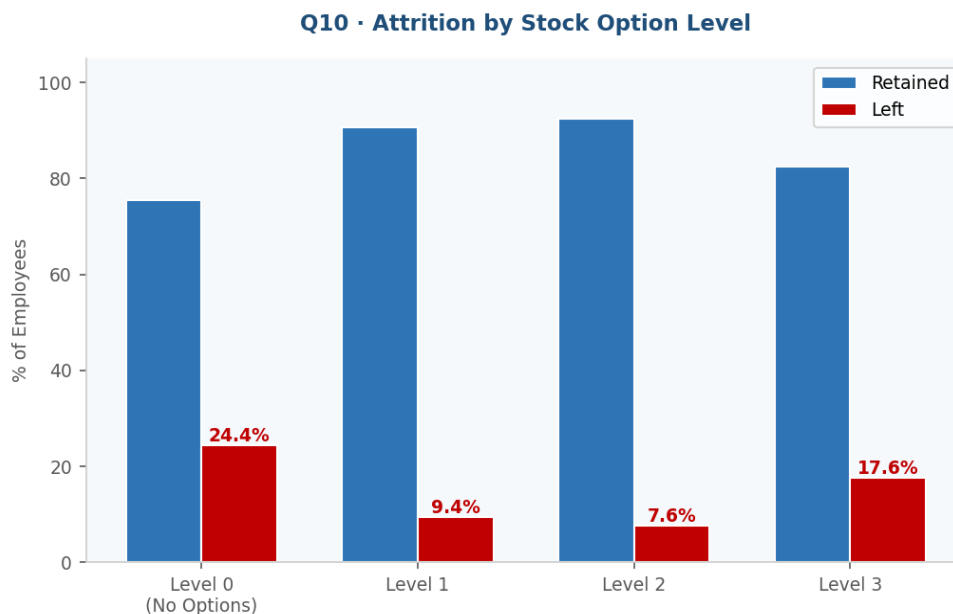
The average salary hike percentage is virtually identical for employees who left (15.1%) and those who stayed (15.2%). The 0.1 percentage point difference is statistically negligible, indicating no meaningful relationship between salary hike percentage and the decision to leave.

Insight:

A salary increase alone does not stop employees from leaving. Even when the percentage increase is the same, lower-paid workers get much smaller actual raises, so it doesn't feel meaningful to them. This also shows that things like workload, job satisfaction, travel, and career growth matter more than small pay increases when deciding to stay or leave.

Recommendation: Focus less on percentage salary increases and more on actual take-home pay. Ensure frontline employees earn a competitive market salary, even if the percentage raise differs.

Question 10: Stock Option Level and Employee Retention



Findings

Stock Option Level	Attrition Rate	Retention Rate
Level 0 (No options)	24.4%	75.6%
Level 1	9.4%	90.6%
Level 2	7.6%	92.4%
Level 3	17.6%	82.4%

Employees with no stock options (Level 0) have a 24.4% attrition rate, dropping sharply to 9.4% at Level 1 and 7.6% at Level 2. Level 3 shows an elevated 17.6%, though this group is small in size and may reflect senior employees leaving for executive opportunities elsewhere.

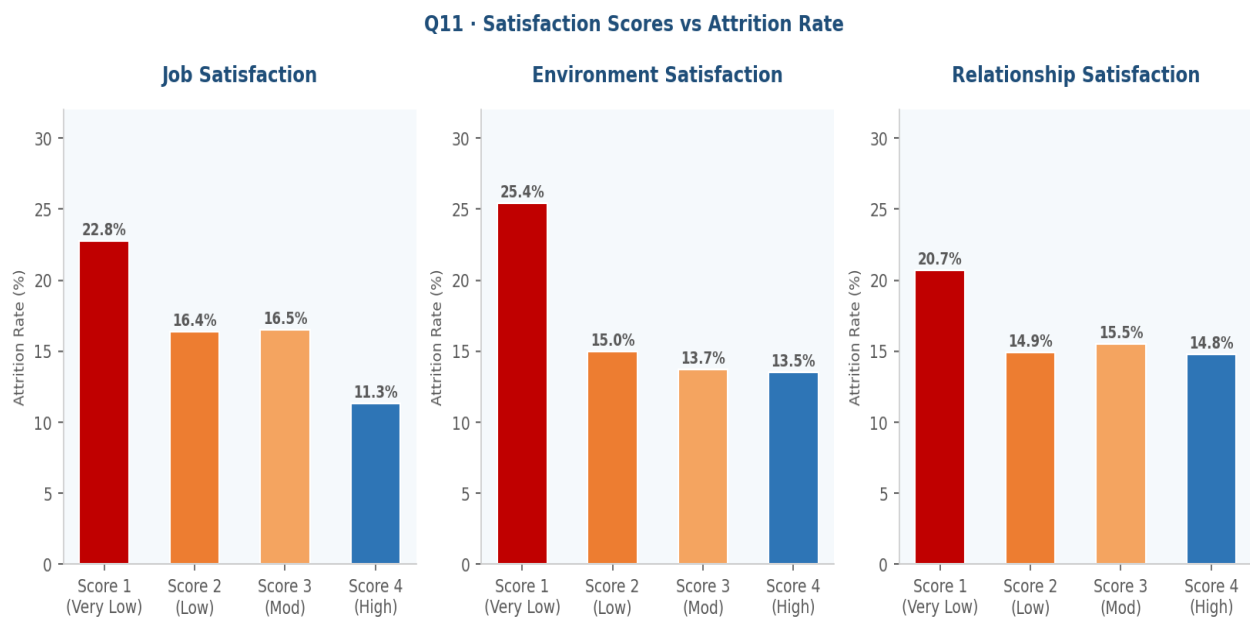
Insight:

Stock options are a strong way to keep employees. Attrition drops by about 15 percentage points when employees move from no stock options to having them.

This happens because employees are less likely to leave when part of their compensation is tied to future company performance.

Recommendation: Extend stock option eligibility to entry-level and mid-level employees. Even a small Level 1 grant can roughly halve attrition rates, making this one of the most cost-effective retention strategies available to TechCrush.

Question 11: Job, Environment, and Relationship Satisfaction vs Attrition



Findings

Satisfaction Metric	Avg Score (Active)	Avg Score (Left)	Gap
Job Satisfaction	2.78	2.47	-0.31
Environment Satisfaction	2.77	2.46	-0.31
Relationship Satisfaction	2.73	2.60	-0.13

Score	Job Satisfaction Attrition	Environment Satisfaction Attrition	Relationship Satisfaction Attrition
1 (Very Low)	22.8%	25.4%	20.7%
2 (Low)	16.4%	15.0%	14.9%
3 (Moderate)	16.5%	13.7%	15.5%
4 (High)	11.3%	13.5%	14.8%

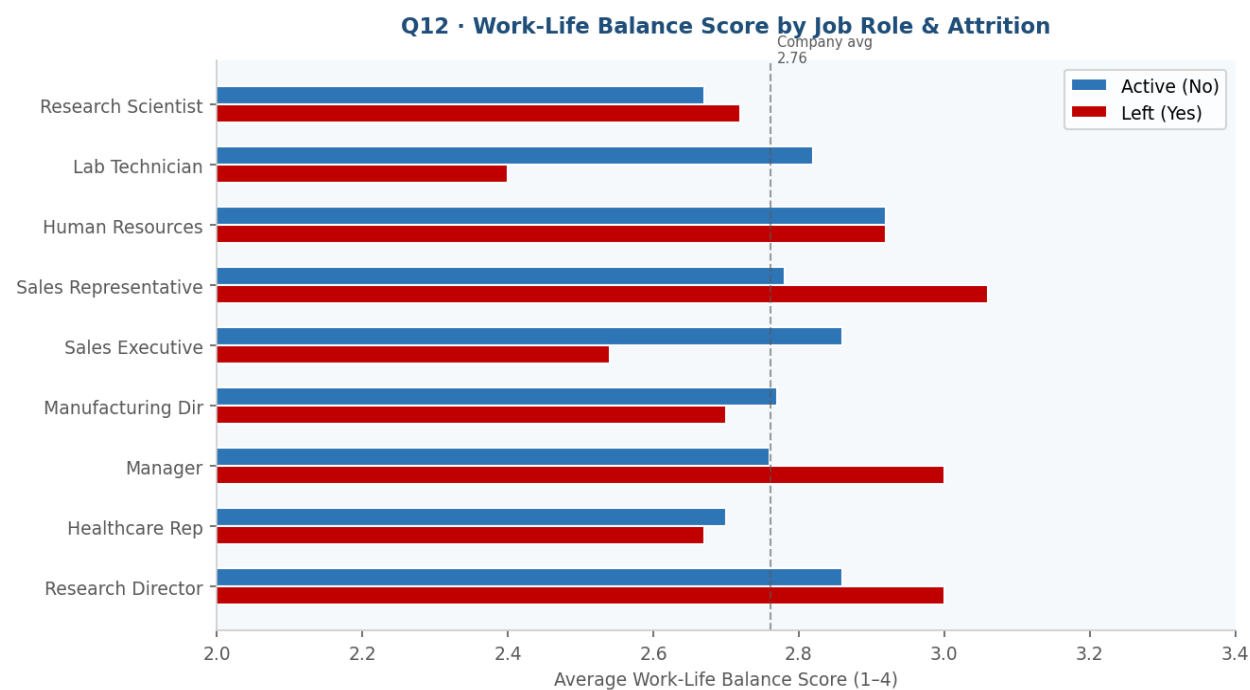
Employees who left consistently reported lower satisfaction across all areas. The biggest difference is in environment satisfaction: those who rated it “1” had a 25.4% attrition rate compared to 13.5% for a rating of “4”. Job satisfaction shows a similar pattern (22.8% vs 11.3%), while relationship satisfaction has a weaker but still noticeable link to attrition.

Insight:

All the three satisfaction areas are linked to whether employees stay or leave. Environment Satisfaction has the biggest impact, low scores nearly double the chance of leaving. Job Satisfaction also strongly affects retention. Relationship Satisfaction matters too, but it is a weaker driver compared to the other two.

Recommendation: The company should run a short quarterly surveys to measure environment, job, and relationship satisfaction by department and role. And then Use the results to quickly identify low-scoring areas (below 2.5) and take action such as improving the work environment, training managers, or redesigning jobs to boost engagement.

Question 12: Work-Life Balance by Job Role and Attrition



Findings

Job Role	WLB Score (Active)	WLB Score (Left)	Attrition Rate
Laboratory Technician	2.82	2.40	23.9%
Sales Executive	2.86	2.54	17.5%
Research Scientist	2.67	2.72	16.1%
Healthcare Representative	2.70	2.67	6.9%
Manufacturing Director	2.77	2.70	6.9%
Sales Representative	2.78	3.06	39.8%
Human Resources	2.92	2.92	23.1%
Manager	2.76	3.00	~5.4%
Research Director	2.86	3.00	2.5%

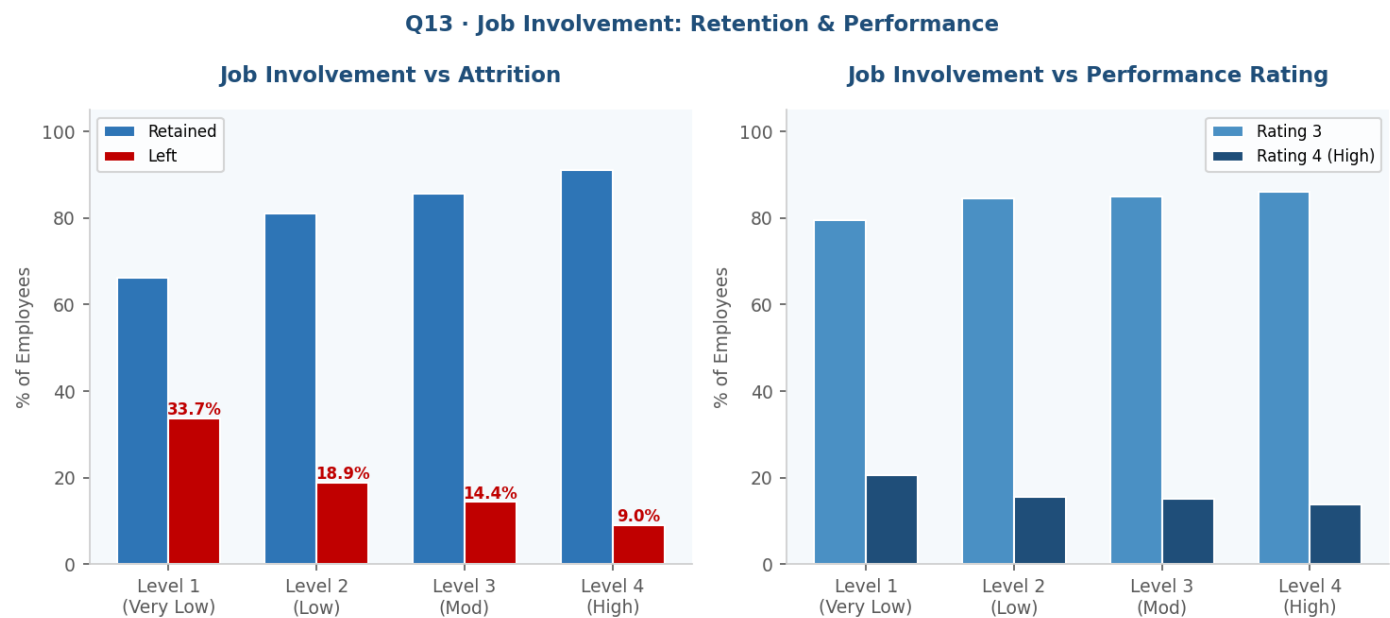
Laboratory Technicians who left had a notably lower WLB score (2.40) than active ones (2.82) — a difference of 0.42 points, this confirmed poor work-life balance as a driver of their high 23.9% attrition rate. Sales Executives who left also scored lower (2.54) than those who stayed (2.86). Interestingly, Sales Representatives who left scored slightly higher on WLB (3.06), indicating that for this role, work-life balance is not the primary reason for leaving.

Insight:

Work-life balance affects roles differently. For Laboratory Technicians, poor WLB strongly contributes to leaving. For Sales Representatives, pay and travel are bigger issues. This means one general WLB solution won't work, each role needs specific fixes based on what drives their stress

Recommendation: For Laboratory Technicians, fix workload, overtime, and schedules. For Sales staff, improve pay and reduce travel stress, since those matter more than work-life balance

Question 13: Job Involvement, Retention, and Performance Ratings



Job Involvement and Attrition

Job Involvement Level	Attrition Rate	Retention Rate
1 (Very Low)	33.7%	66.3%
2 (Low)	18.9%	81.1%
3 (Moderate)	14.4%	85.6%
4 (High)	9.0%	91.0%

Job Involvement and Performance Rating

Job Involvement Level	% with Rating 3	% with Rating 4 (High)
1 (Very Low)	79.5%	20.5%
2 (Low)	84.5%	15.5%
3 (Moderate)	84.9%	15.1%
4 (High)	86.1%	13.9%

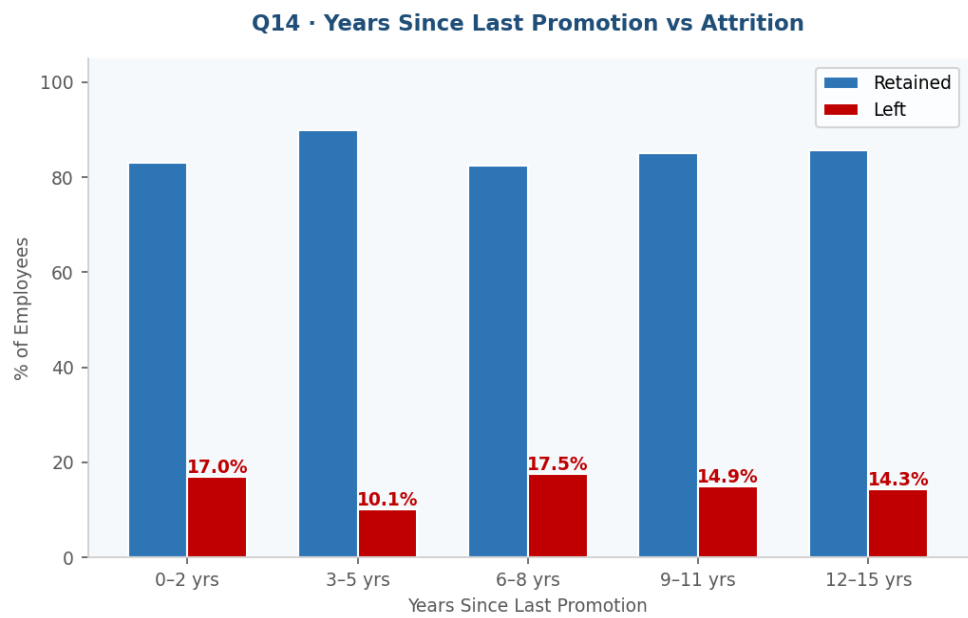
Employees with the lowest job involvement (Level 1) have a 33.7% attrition rate that is about 4x the 9.0% rate of highly involved employees (Level 4). However, performance ratings are remarkably similar across all involvement levels, with most employees receiving a rating of 3 regardless of involvement.

Insight:

Job involvement strongly affects whether employees stay, but not how they are rated at work. This means some employees may still perform well but are already thinking of leaving. So, involvement scores should be used to spot early signs of potential turnover, not to judge performance

Recommendation: Focus on employees with low job involvement (scores 1–2). Have early career conversations with them and offer more engaging tasks, growth opportunities, and mentorship to keep them from leaving.

Question 14: Years Since Last Promotion and Attrition



Findings

Years Since Last Promotion	Employees (Total)	Employees Left	Attrition Rate
0–2 years	1,097	186	17.0%
3–5 years	158	16	10.1%
6–8 years	126	22	17.5%
9–11 years	47	7	14.9%
12–15 years	42	6	14.3%

Attrition is highest among employees promoted within the last 2 years (17.0%) and those who have gone 6–8 years without promotion (17.5%). The lowest attrition is at 3–5 years post-promotion (10.1%), suggesting a post-promotion satisfaction'. The 0–2 year group also accounts for the majority of all attrition in absolute numbers (186 of 237 total leavers).

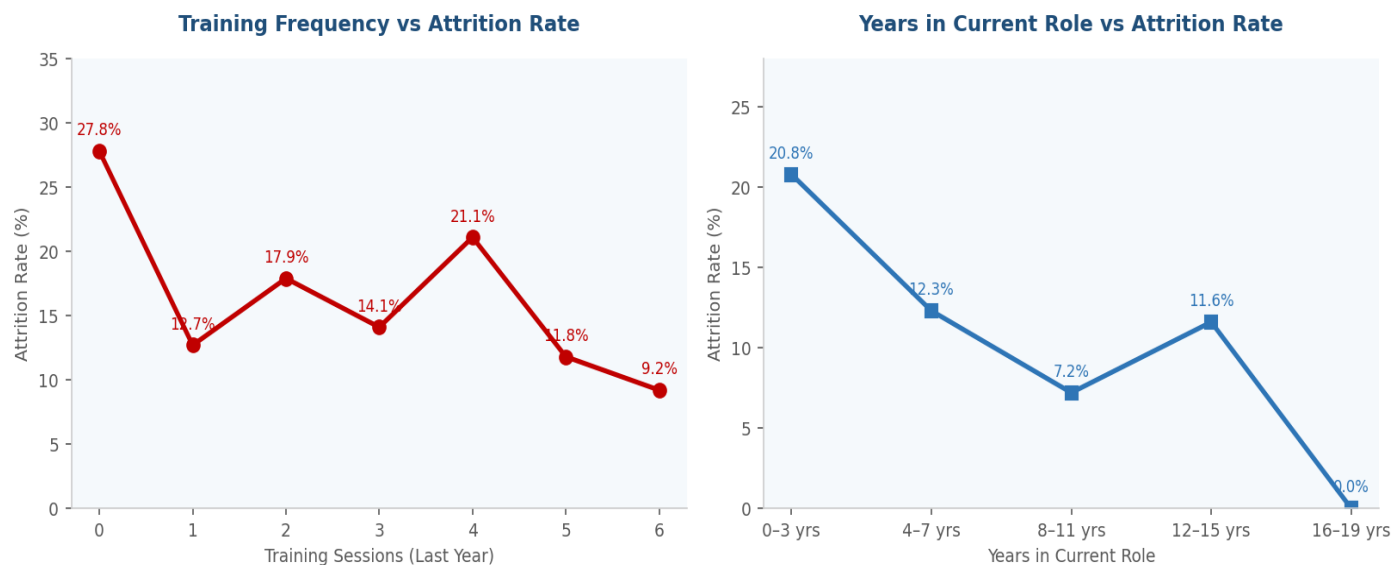
Insight:

Two clear risk groups stand out here: First, employees who were recently promoted (0–2 years) are leaving the most, they are likely still deciding if the role meets their expectations. Second, those with 6–8 years in the company without promotion also leave a lot due to feeling stagnant. And possibly after 3–5 years, if there’s no next promotion, dissatisfaction builds and people start to exit.

Recommendation: Techcrush should make career growth paths clear and easy to understand. Talk to employees who haven't been promoted for about 5 years, and help them plan their next step. Also, make promotion requirements transparent and realistic.

Question 15: Training Frequency, Years in Current Role, and Performance

Q15 • Training & Role Tenure vs Attrition



Training Frequency vs Attrition

Training Sessions	Attrition Rate	Avg Perf (Active)	Avg Perf (Left)
0 sessions	27.8%	3.15	3.27
1 session	12.7%	3.15	3.00
2 sessions	17.9%	3.16	3.19
3 sessions	14.1%	3.16	3.09
4 sessions	21.1%	3.12	3.15
5 sessions	11.8%	3.14	3.29
6 sessions	9.2%	3.15	3.00

Years in Current Role vs Attrition

Years in Role	Attrition Rate	Avg Perf (Active)	Avg Perf (Left)
0–3 years	20.8%	3.14	3.15
4–7 years	12.3%	3.16	3.18
8–11 years	7.2%	3.17	3.13
12–15 years	11.6%	3.29	3.00

Years in Role	Attrition Rate	Avg Perf (Active)	Avg Perf (Left)
16–19 years	0.0%	3.15	N/A

Employees who received no training had the highest attrition at 27.8%, falling to 9.2% for those with 6 training sessions. Attrition also decreases steadily with years in role, from 20.8% (0–3 years) down to 7.2% (8–11 years). However, performance ratings remain essentially close across all training levels and role tenures (ranging closely from 3.12 to 3.29), indicating that neither training frequency nor years significantly shifts measured performance.

Insight:

Training strongly reduces attrition, employees with no training leave almost 3 times more than those with 6 trainings. This is likely because training makes employees feel valued and invested in.

Performance ratings don't change much across employees, suggesting the rating system may not be detailed enough. Also, longer role tenure lowers attrition up to about 11 years, after which it becomes stable.

Recommendation: Techcrush should make sure every employee gets at least one training each year, especially those who haven't had any recently. And treat training as a way to show employees they are valued. Also, improve the performance rating system so it can better distinguish high and low performers for fair rewards.

Conclusion

TechCrush is facing a clear employee retention problem, with an attrition rate of 16.1%. This means the company is losing important mid-level employees, which can affect experience, teamwork, and overall performance. However, the good news about all of these is that the problem is not random, it has clear causes like overtime, pay gaps, lack of training, and limited stock options. These are issues the company can actually fix.

By addressing these areas together, TechCrush can reduce attrition within about 12–18 months, and to keep improving, the company should regularly track HR data using dashboards, run short employee surveys, and review turnover trends every year so they can quickly be aware of possible problems and respond to them effectively.